

When design predicts disease: residualisation failure and generalisation gaps in donor-level single-cell classifiers

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Abstract

Single-cell RNA sequencing can resolve disease-associated immune states at cellular resolution, but clinical classification is performed at the patient level. A rapidly growing literature therefore converts thousands of cells into one donor representation and reports cross-validated discrimination, often above ROC-AUC 0.95. Whether these values reflect disease biology, study design, or a mixture of both is rarely tested.

We reconstructed donor-level design metadata for two widely reused systemic lupus erythematosus (SLE) cohorts and evaluated frozen Geneformer donor means, highly variable gene (HVG) pseudobulk and PCA pseudobulk under one locked donor-level pipeline. In GSE174188, processing-wave composition alone predicted disease at AUC 0.922, while in GSE135779 the complete measured design, sequencing and demographic block reached 0.952. Every representation recovered recorded batch identity with AUC up to 0.9997. Yet the inference that internal discrimination was merely a batch shortcut was contradicted by design restriction: limiting GSE174188 to dominant or pure wave 4 reduced AUC by only 0.014-0.049 relative to size- and label-matched donor subsets, and removing the all-case B1 batch from GSE135779 changed AUC by -0.012 to +0.003.

Conventional residualisation gave the opposite answer. Removing processing-wave coordinates in GSE174188 reduced representation AUC by 0.230-0.268. In GSE135779, residualising batch alone changed almost nothing, whereas projecting out the complete design block collapsed all three representations to AUC 0.515-0.526. The magnitude of this collapse tracked how accurately the projected covariates predicted the outcome, not how much performance disappeared under design restriction. Residualisation therefore cannot identify technical confounding when design and disease are collinear; it removes disease-aligned signal by construction.

Independent transfer exposed the failure that internal adjustment could not resolve. Moving across processing waves reduced AUC to 0.806-0.926, and strict source-only cross-cohort transfer favoured expression pseudobulk over frozen Geneformer in the 261-donor target. Three learned set-pooling architectures did not recover the gap. We introduce an executable validity standard for patient-level single-cell classification: report design-only AUC, representation-to-design predictability, design-restricted performance with matched donor controls, and source-only external performance. High internal AUC is evidence of separability; only this chain tests whether the separation is attributable and transportable.

Keywords: patient-level classification; single-cell RNA sequencing; study design; batch confounding; residualisation; external validation; Geneformer; pseudobulk; systemic lupus erythematosus

Author Summary

Single-cell RNA sequencing measures thousands of cells from each patient, but many clinical studies ultimately predict one outcome per patient. A model can therefore appear highly accurate because it captures disease biology, differences in how samples were collected and processed, or both. We reconstructed sample-design information for two public lupus cohorts and tested three common patient representations: frozen Geneformer embeddings and two expression-based pseudobulk summaries. Study-design variables alone predicted disease with high accuracy, and all three molecular representations identified processing batches almost perfectly. However, two standard attempts to diagnose this problem disagreed. Comparing patients within overlapping batches largely preserved disease discrimination, whereas mathematically removing design-associated components sometimes erased most

of it. The latter operation cannot distinguish technical bias from real biology when study design and disease status are closely aligned. Independent transfer between batches and cohorts exposed a more consistent loss of generalisation, and learned pooling did not repair it. We propose four inexpensive checks that patient-level single-cell classifiers should report: disease prediction from design alone, design prediction from the representation, analysis within overlapping designs with matched controls, and fully source-fitted external validation.

1. Introduction

Systemic lupus erythematosus is a heterogeneous autoimmune disease in which blood interferon activity and immune-cell states vary substantially between patients. Single-cell RNA sequencing can resolve those states, but diagnosis concerns a patient while the measurements are made in individual cells. Patient-level modelling must therefore compress a variable-sized cell population into one representation without confusing biological signal with the way the cohort was assembled.

Several solutions now compete for this role. Pseudobulk aggregates expression, CloudPred and singleDeep learn phenotype-oriented summaries, and recent methods such as PaSCient, scPhase and FloREN learn multicellular or graph-based patient representations [6-11]. Frozen single-cell foundation models offer a particularly attractive shortcut: encode each cell once, average the embeddings, and train a small donor classifier. This strategy is computationally convenient, but high within-cohort accuracy does not establish that the representation will survive a change of hospital, population, acquisition period or sequencing workflow.

The hidden difficulty is that many public disease cohorts are not random samples from one acquisition process. Cases and controls may come from different recruitment sources, years, multiplexing pools or processing waves. Batch-label confounding has long been known to bias cross-validation [16], and a completely confounded single-cell design is statistically non-identifiable [17]. Nevertheless, most patient-level single-cell studies treat this problem as a preprocessing question: regress a representation on technical covariates and re-evaluate the residuals. When those covariates are also proxies for disease, residualisation removes the disease-aligned component whether it is technical or biological [18,19]. A lower post-residualisation AUC can therefore be mistaken for proof that the original model used a shortcut.

This problem has become more urgent, not less. Broad benchmarks show that frozen single-cell foundation-model embeddings can retain batch structure and can lose to simple expression baselines [13,14]. Patient-level methods already report high internal performance in SLE, including external validation between GSE174188 and GSE135779 [7]. FloREN now evaluates both cohorts with supervised graph representations [11], while pretraining-contamination work argues that benchmark validity must accompany model rank [15]. What remains missing is a direct empirical test of whether design predictability, residualisation, design restriction and external transfer tell the same story on the same donors. They do not.

Here we recover donor design metadata for GSE135779 and combine it with the cell-distributed processing metadata in GSE174188. Across frozen Geneformer and two expression pseudobulk representations, we place design-only prediction, representation-to-design prediction, fold-contained residualisation, design-restricted evaluation, matched donor controls, cross-batch transfer, cross-cohort transfer and learned pooling under one donor-matched framework. To our knowledge, this is the first patient-level single-cell study to quantify this complete validity chain across two disease cohorts. The result is not another leaderboard. It is a demonstration that separability, adjustment and transportability are different claims and require different experiments.

2. Results

2.1. Study design predicts disease in two widely reused SLE cohorts

GSE174188 contains 261 analysed CD4-positive donors: 162 SLE cases and 99 controls [2]. The distributed single-cell object records four processing waves. In the CD4 subset, their control/case counts were 32/0, 17/97, 7/19 and 43/46 (Figure 1A). Wave 1 contains no cases, and wave 2 is dominated by cases. A logistic model using

only donor processing-wave fractions achieved AUC 0.922 (2.5th-97.5th percentile over 20 repeated five-fold donor analyses: 0.916-0.928). Sequencing/QC variables reached 0.781, and the complete measured design block reached 0.953 (Table 1; Figure 1D).

For GSE135779, we restored batch, collection year, demographics, clinical fields and sequencing QC for all 44 childhood-cohort donors from the original Supplementary Tables 1b and 1c and mapped them to the analysis identifiers through GEO [1]. No donor or selected field was missing. The six sequencing batches contained case/control counts of 8/0, 3/1, 6/2, 6/2, 6/2 and 4/4 (Figure 1B). Collection years 2014 and 2016 contained only cases, whereas 2018 was balanced at 8/8 (Figure 1C). Collection year alone predicted disease at AUC 0.721, demographics at 0.715 and sequencing/QC at 0.707. Their combination with batch reached AUC 0.952 (0.893-0.992).

These two cohorts encode design-label entanglement differently. GSE174188 has one dominant processing coordinate that is itself a strong disease classifier. GSE135779 distributes the label channel across year, sequencing and demographic variables. A batch table alone would expose the first design but miss the second.

Table 1. Cohort design structure and design-only discrimination.

Cohort	Donors (case/control)	Recorded design	Complete-design AUC	Most informative single block
GSE135779	44 (33/11)	6 batches; 4 collection years; sequencing QC; demographics	0.952 [0.893-0.992]	collection year: 0.721
GSE174188 CD4	261 (162/99)	4 processing-wave fractions; sequencing QC; age; sex	0.953 [0.945-0.961]	processing wave: 0.922
GSE285773 CD4	26 (16/10)	donor labels and cell profiles; no recoverable donor processing batch	not estimated	not estimated

2.2. The representations remember the design more accurately than the disease

We next asked whether the donor representations retained the recorded design identity. Under the same fold-contained pipeline, each representation predicted case/control status and every batch one-versus-rest (Figure 2).

In GSE174188, Geneformer, HVG pseudobulk and PCA pseudobulk classified disease at AUC 0.982, 0.977 and 0.975. Their best processing-wave AUCs were 0.9997, 0.999 and 0.997. In GSE135779, disease AUCs were 0.870, 0.933 and 0.945, while the best individual-batch AUCs were 0.993, 0.961 and 0.961. Near-perfect batch recognition was therefore not a peculiarity of one representation family or one cohort.

This result establishes exposure, not attribution. A representation that predicts batch can still contain disease information, and a batch-aware classifier need not use batch as its disease decision rule. The distinction is resolved by restriction, not by correlation alone.

2.3. Disease discrimination survives design restriction

We first restricted GSE174188 to dominant wave 4, the only large near-balanced wave (n=89; 46 cases and 43 controls). Because reducing 261 donors to 89 inevitably lowers performance, we compared the observed stratum with 20 random donor subsets having the same donor count and case/control composition (Figure 3A). Relative to those matched subsets, AUC changed by -0.014 for Geneformer, -0.028 for HVG pseudobulk and -0.049 for PCA pseudobulk.

The stricter analysis retained only donors with at least 99% of analysed cells in wave 4 (n=66; 25 cases and 41 controls). Relative to size- and label-matched random subsets, AUC changed by -0.026, -0.044 and -0.044 (Figure

3B). These are detectable losses, but they are far smaller than the raw full-cohort-to-stratum difference. Most of that raw difference is explained by using fewer donors.

GSE135779 provided an independent design. Removing the all-case B1 batch left 36 donors from batches B2-B6, with both labels represented in every retained batch. Geneformer AUC was 0.846 versus 0.858 in matched random subsets (difference -0.012). Both pseudobulk representations were 0.919 versus 0.916 (difference +0.003; Figure 3C). Thus the disease signal persisted after the most obvious label-exclusive batch was removed.

Across both cohorts, design restriction weakens the claim that internal AUC is a pure technical shortcut. The representations carry design information, but the measured disease separation does not disappear when the clearest design channels are held fixed.

2.4. Residualisation can erase the signal that restriction preserves

Residualisation produced a sharply different conclusion. All unadjusted and adjusted models were recomputed on identical outer folds with the same training-only HVG selection, scaling, PCA construction, regularisation grid and classifier.

In GSE174188, ridge residualisation on processing-wave fractions reduced Geneformer AUC from 0.982 to 0.714, HVG pseudobulk from 0.977 to 0.747 and PCA pseudobulk from 0.975 to 0.735 (losses 0.268, 0.230 and 0.241; Figure 4A). The same design, tested by dominant- or pure-wave restriction with matched donor controls, cost only 0.014-0.049. Residualisation therefore removed five to nineteen times more AUC than the matched restriction discrepancy, depending on representation and stratum.

GSE135779 revealed why. Batch alone had a cross-validated disease AUC of 0.499, and batch residualisation changed representation AUC by +0.002, -0.002 and -0.009 (Figure 4B). Collection year had AUC 0.721 and its removal cost 0.005-0.039. The complete measured design block had AUC 0.952; projecting it out collapsed Geneformer, HVG pseudobulk and PCA pseudobulk to 0.516, 0.526 and 0.515, removing 0.354, 0.407 and 0.430 AUC (Figure 4C-D). Yet exclusion of the all-case batch had changed matched performance by at most 0.012.

The projected covariate block predicts how much discrimination residualisation will erase. This is expected algebraically: when design predicts the label, the component of a representation predictable from design includes disease-aligned variation. Residualisation cannot determine whether that variation is technical, biological or both. It answers how much discrimination survives deleting design-aligned signal, not how much of the original discrimination was caused by technical batch.

This empirical discrepancy is the paper's central finding. A large post-residualisation loss is not evidence that the original AUC was spurious. Without overlap-stratum evaluation, it is equally compatible with successful removal of a shortcut and destructive removal of outcome-aligned biology.

2.5. Generalisation, rather than internal separability, is the reproducible failure

Design restriction preserved strong within-cohort discrimination, but transfer weakened it at every level tested (Figure 5). Training on processing waves 2+3 and testing on wave 4 in GSE174188 produced AUC 0.842, 0.811 and 0.806 for Geneformer, HVG pseudobulk and PCA pseudobulk. Reversing the direction produced 0.919, 0.926 and 0.821. Wave 1 could not be evaluated as a target because it contained no cases.

Strict source-only transfer from the 26-donor GSE285773 cohort to the independent 261-donor GSE174188 target yielded AUC 0.884 [0.843-0.920] for Geneformer, 0.919 [0.884-0.946] for HVG pseudobulk and 0.926 [0.895-0.953] for PCA pseudobulk. On the same 261 target donors, both pseudobulk methods exceeded Geneformer after paired DeLong testing and Benjamini-Hochberg adjustment ($q=0.0151$ and $q=0.000603$). The reverse direction retained the same ranking but was underpowered at 26 target donors.

Restricting the GSE174188 source to dominant wave 4 did not improve transfer to GSE285773. Target AUC fell to 0.794, 0.894 and 0.888 and was no better than size- and label-matched random source subsets. More varied source donors transferred better than the apparently cleaner restricted source.

The cross-cohort result has a direct translational interpretation without implying clinical readiness: a molecular classifier is useful beyond its development cohort only if patient ranking survives changes in age distribution, recruitment and technical context. Internal AUC did not predict that property here.

2.6. Learned pooling does not repair the transfer gap

Mean pooling might suppress informative cell heterogeneity, so we tested DeepSets, gated-attention multiple-instance learning and pooling by multi-head attention [20-22]. All operators used the same frozen cell embeddings, cap-500 cells per donor, a source-fitted whitened PCA32 projection and source-only hyperparameter selection. Independent target data were never used to select a model (Figure 6).

In the 261-donor target, none of the learned operators beat ordinary mean pooling. Relative to the 1,152-dimensional Geneformer donor mean, their paired AUC differences were -0.133, -0.055 and -0.042. Relative to a donor mean in the identical PCA32 space, the differences were -0.077, +0.001 and +0.014; none of the positive differences was significant. All learned operators remained below both pseudobulk representations. In the 26-donor target, no learned-pooling comparison was significant.

This is a useful negative method result. The external gap was not repaired by adding permutation-invariant capacity to the frozen embedding. A more expressive pooling function can fit a source cohort without creating a transportable patient representation.

2.7. An executable validity standard

The results support four minimum requirements for patient-level single-cell classification:

- **Report design-only disease AUC.** Batch, year, centre, cell yield, sequencing

quality and demographic variables should be evaluated as a label-only baseline.

- **Report representation-to-design predictability.** A representation that retains

design identity at near-perfect AUC has documented exposure to cohort structure, even if attribution remains unresolved.

- **Pair residualisation with design restriction and matched donor controls.**

Residualisation alone is uninterpretable under label-design collinearity.

- **Require a source-only external target.** Target labels must not choose feature

transformations, and internal performance must be reported beside external performance.

These checks are inexpensive relative to embedding generation or deep pooling. They convert “the model achieved high AUC” into four separable statements: the label is predictable, the design is visible, the signal survives observed overlap, and the ranking transfers.

3. Discussion

This study identifies a validity failure that representation leaderboards do not measure. In two widely reused SLE cohorts, measured study design can classify disease almost as accurately as the molecular representations, and every representation can recover batch identity with near-perfect accuracy. That combination should trigger a confounding investigation. It does not, however, tell us whether the disease classifier is a batch classifier.

The decisive comparison is between two adjustment strategies. Restriction asks whether discrimination survives among donors observed under the same or overlapping design. Residualisation deletes every coordinate predictable from a chosen design block. When design and disease are collinear, these operations target different quantities. The two-cohort result makes the distinction visible: mixed-batch restriction preserved GSE135779 performance, while residualising the complete design block reduced every representation to chance. In GSE174188, wave restriction caused modest matched losses, while wave residualisation removed roughly one

quarter of the AUC scale.

The result is more consequential than a contest between Geneformer and pseudobulk. Recent benchmarks already show that simple expression baselines can outperform frozen foundation-model embeddings [13,14], and patient-level representation learning is an active field [8-11]. Our contribution is to show why an internal comparison can remain scientifically ambiguous even when donor leakage is prevented, folds are strict and adjustment is performed inside each fold. Correct computation does not repair a non-identifiable design.

The external results identify what can be concluded. Strong discrimination persisted inside design-restricted strata, supporting a reproducible disease-aligned signal. The same representations lost performance across processing waves and cohorts, showing that the signal was not sufficiently invariant. Pseudobulk converted additional source donors into transferable information more effectively than frozen Geneformer in the evaluated CD4 setting, and learned pooling did not reverse the ordering. These are not contradictory findings: internal separability can be real while external generalisation remains inadequate.

For clinical translation, this reframes representation selection as a cross-centre validity problem. A classifier that ranks patients inside one retrospective cohort but changes under a new acquisition design is not ready for probability interpretation or individual decisions. The present external AUCs quantify ranking, not calibrated risk. The immediate deliverable is therefore not a diagnostic model but a validation protocol that future diagnostic models should pass.

4. Limitations

This study is confined to three public SLE cohorts, frozen Geneformer final-layer CLS embeddings, fixed expression baselines and a small set of learned pooling operators. The overlap strata are smaller than the full cohorts, and matched random subsets separate sample-size loss from design restriction without identifying a causal technical effect. Recorded batch can combine recruitment source, age, ancestry, disease severity and laboratory processing; our result is non-attributability, not proof that the retained signal is purely biological. GSE285773 lacks recoverable donor processing-batch metadata. External bootstrap intervals condition on one fitted source model, the 26-donor target has limited power, and external probabilities are not clinically calibrated. Geneformer pretraining overlap with individual benchmark cells was not tested. These boundaries do not weaken the central empirical finding: residualisation, restriction and transfer produced different answers on the same donors and cannot substitute for one another.

5. Methods

5.1. Cohorts, outcomes and statistical unit

We analysed GSE135779 (44 childhood donors; 33 SLE, 11 controls), the CD4-positive alpha-beta T-cell subset of GSE174188 (261 donors; 162 SLE, 99 controls) and GSE285773 CD4-positive T cells (26 donors; 16 SLE, 10 controls) [1-3]. The donor was the independent unit in every supervised analysis. Cells from one donor never crossed training and test partitions.

5.2. Recovery of GSE135779 design metadata

GSE135779 batch, collection year, age, sex, race, ethnicity and clinical variables were read from the original Supplementary Table 1b; per-library sequencing metrics were read from Supplementary Table 1c [1]. Study names were linked to analysis donor identifiers through the GEO title/accession map. All 44 donors mapped uniquely. The restoration script writes the donor table, batch-by-label, year-by-label and batch-by-year cross-tabs, and SHA256 hashes of every source file.

For GSE174188, per-cell `Processing_Cohort` was read from the distributed CELLxGENE object and reduced to donor fractions over four waves. Dominant wave was the maximum-fraction wave; a pure-wave donor had at least 99% of analysed cells in one wave.

5.3. Frozen Geneformer embeddings

Cell embeddings used Geneformer V2-316M at repository revision

04c2b2e84da7c0f385c3f9ad8f3ec24bab6650e5 [4]. The archived checkpoint, configuration, token dictionary and gene-median dictionary SHA256 hashes are provided in the methods manifest. Gene identifiers were version-trimmed and mapped to the V2 vocabulary. Within each cell, the ranking value was $\text{raw count} / \text{total count} * 10,000 / \text{Genecorpus-104M gene median}$; genes were sorted descending. Sequences used V2 start/end token IDs 2/3, padding ID 0 and maximum length 4096, leaving at most 4094 ranked genes.

The model output was the final `last_hidden_state`; token position 0 was retained as the 1,152-dimensional cell CLS vector. The encoder was frozen. Donor representations were coordinate-wise means of sampled cell vectors. Sampling without replacement used integer seed 1 and caps of 500 cells per donor for GSE135779 and 1,000 for GSE174188 and GSE285773. The embedding environment used Python 3.10, PyTorch 2.5.1 and Transformers 4.46.3.

5.4. Pseudobulk representations

Stored donor-by-gene values were $\log_{1p}(\text{CPM})$. Within every outer training fold, gene variance was computed on training donors, the 4,000 highest-variance genes were selected, and coordinates were standardised by training-donor mean and standard deviation. HVG pseudobulk used these values directly. PCA pseudobulk fitted up to 30 components on the training values and applied the fitted map to held-out donors.

For cross-cohort transfer, trimmed gene identifiers were intersected before source-only HVG selection. Feature means, variances, selected genes, scaler, PCA, classifier and regularisation were fitted on source donors and applied unchanged to the target.

5.5. Repeated donor-level evaluation

Internal analyses used 20 repeated stratified five-fold donor splits with integer seeds 20260801-20260820. The classifier was balanced logistic regression with the liblinear solver and `max_iter=20,000`. Inverse regularisation strength was selected from 10^{-4} through 10^4 in decade steps using five-fold resampling within each outer training set. The full outer-training representation was used for this inner regularisation selection; no outer test donor entered HVG selection, scaling, PCA or classifier fitting.

Out-of-fold donor predictions were pooled within each repeat. We report mean AUC and the 2.5th-97.5th percentile of repeat-level AUCs. This is a split-sensitivity range, not a population confidence interval.

5.6. Design-only and representation-to-design prediction

Numeric design variables were median-imputed and standardised inside the outer training fold. Categorical variables were most-frequent-imputed and one-hot encoded, with held-out unknown levels ignored. The same logistic model and regularisation grid were used. GSE135779 blocks were batch, collection year, sequencing QC, demographics and their union. GSE174188 blocks were processing-wave fractions, sequencing QC, demographics and their union.

Each molecular representation also predicted every observed batch one-versus-rest under five repeated donor-level cross-validations. The number of folds was capped by the minority batch count.

5.7. Fold-contained residualisation

Unadjusted and residualised models shared outer folds, feature construction, scaling, PCA and classifier selection. Within each outer training fold, design variables were encoded as above and a multivariate ridge regression ($\alpha=1$, intercept included) mapped design to every raw representation coordinate. Training and held-out coordinates were replaced by observed minus predicted values, after which the identical representation-specific pipeline was fitted to training residuals and applied to held-out residuals. The disease label was not included in the residualiser.

5.8. Design restriction and matched controls

GSE174188 was restricted to dominant wave 4 and separately to pure wave 4. GSE135779 excluded all-case batch B1 and retained B2-B6, each of which contained both labels. The complete internal pipeline was rerun after restriction.

For every observed stratum, 20 donor subsets were drawn without replacement from the full cohort with the identical donor count and case/control count. Each draw used one prespecified split seed. The observed-minus-matched difference quantifies whether the stratum is harder than expected from reduced donor number and label composition.

5.9. Cross-batch and cross-cohort transfer

Cross-batch analyses trained on waves 2+3 and evaluated wave 4, then reversed the direction. Cross-cohort transfer fitted every statistic on the source cohort. Target labels were accessed only after prediction.

Target AUC intervals used 5,000 case/control-stratified donor bootstrap resamples and the percentile method. Paired AUC comparisons used DeLong tests on the identical target donors [23] and Benjamini-Hochberg correction within declared method families [24].

5.10. Learned pooling

DeepSets, gated-attention multiple-instance learning and pooling by multi-head attention operated on cap-500 frozen cell embeddings [20-22]. Cell standardisation, a whitened source PCA32 projection, network weights and hyperparameters were source-fitted. Hidden widths were 16 or 32, weight decay 0.01 or 0.1, and Adam used a learning rate of 0.02 for 150 epochs. Five-fold source-donor cross-validation selected the configuration; three initialisations were fitted on all source donors and target probabilities were averaged. An ordinary donor mean in the identical PCA32 space isolated the contribution of learned pooling.

The PCA32 projection used for hyperparameter selection was fitted once on all source cells before source-donor folds. It never used target cells or labels. Consequently, source-internal pooling AUC was not used as an unbiased generalisation estimate; only independent-target and paired target-donor comparisons support the learned-pooling conclusion.

5.11. Software and provenance

The locked donor-level reanalysis used Python 3.13.7, NumPy 2.4.6, pandas 3.0.3, SciPy 1.18.0, scikit-learn 1.9.0, PyArrow 21.0.0 and joblib 1.5.3. Input hashes, seeds, parameter grids, donor-level predictions and output manifests accompany the analysis.

6. Data and code availability

The source datasets are publicly available through GEO accessions GSE135779, GSE174188 and GSE285773. The current public benchmark archive is available at <https://github.com/LightChainr/rheumlens> and concept DOI <https://doi.org/10.5281/zenodo.20813922>. The design-validity reconstruction, restored GSE135779 metadata, locked donor predictions and new figure source tables will be deposited as a new major scientific release after final manuscript and author-metadata validation.

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Figure 1 Outcome labels are entangled with study design in two widely reused SLE cohorts

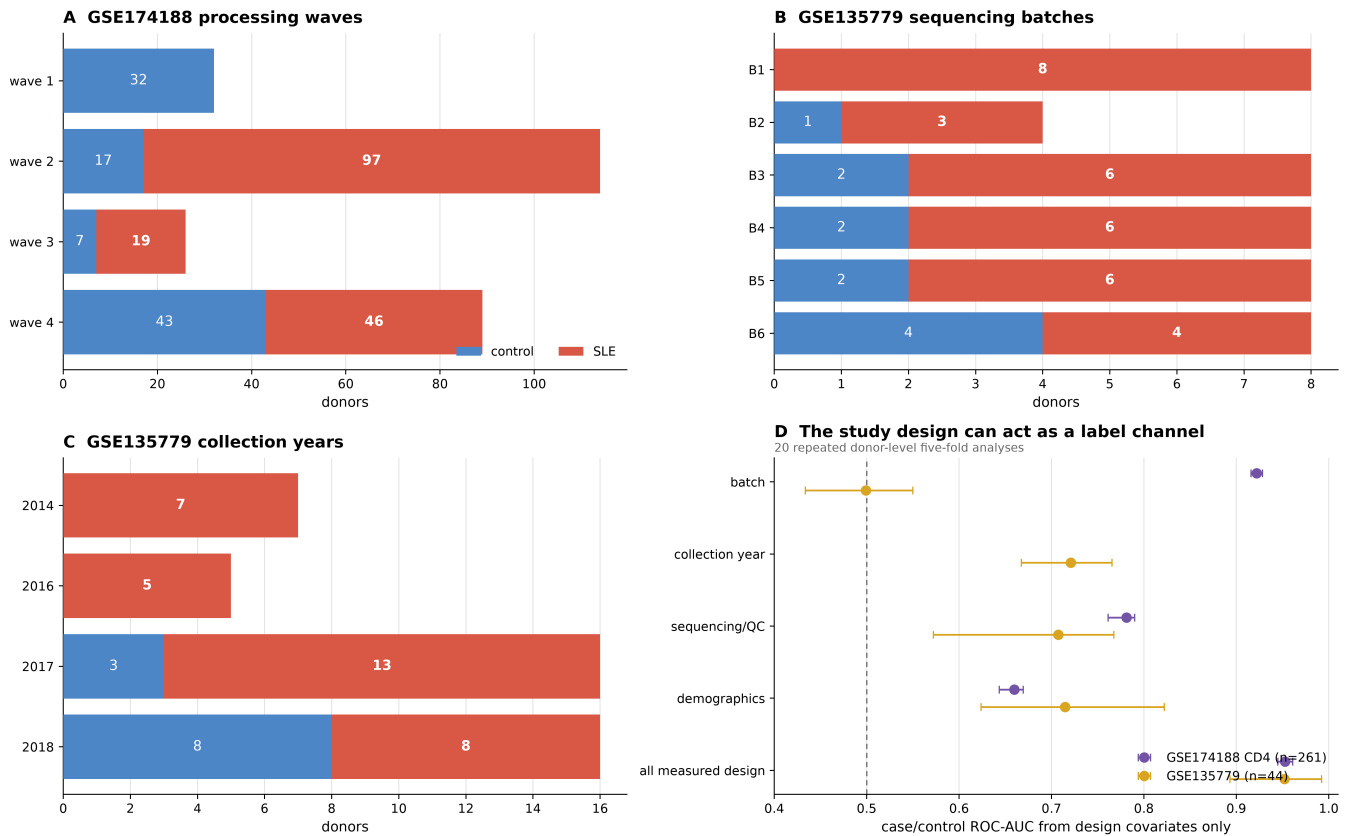


Figure 1. Outcome labels are entangled with study design in two widely reused SLE cohorts (A) Case and control counts across the four processing waves represented among the analysed GSE174188 CD4-positive alpha-beta T-cell donors. Wave 1 contains no SLE cases, whereas wave 2 is dominated by cases. (B) Case and control counts across the six restored GSE135779 sequencing batches. Batch B1 contains only cases; batches B2-B6 contain both labels. (C) Case and control counts by GSE135779 collection year. (D) Repeated donor-level cross-validated ROC-AUC obtained using recorded design variables alone. Points denote mean AUC across 20 repeated stratified five-fold analyses and bars denote the 2.5th-97.5th percentile of repeat-level AUCs. The vertical dashed line marks chance performance. Purple and ochre indicate GSE174188 and GSE135779, respectively.

Figure 2 High disease discrimination coexists with near-perfect design recognition

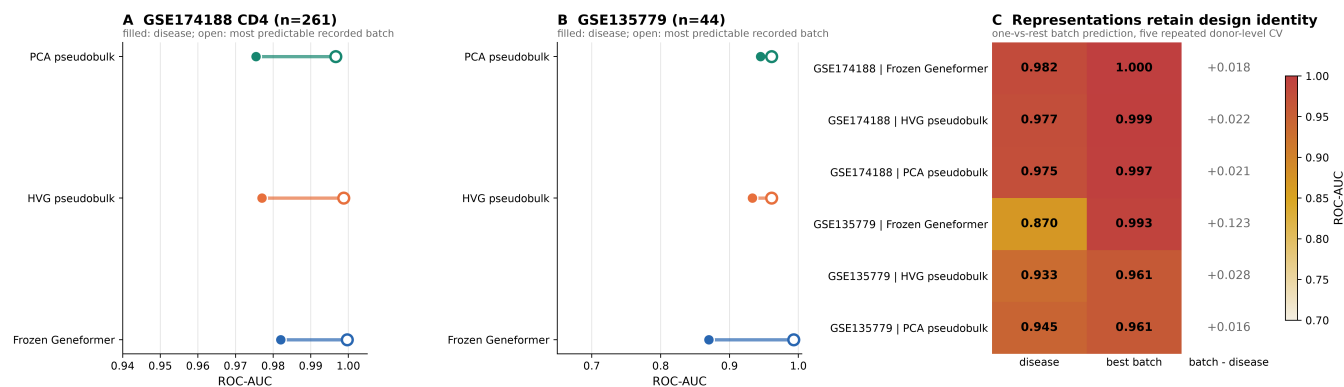


Figure 2. High disease discrimination coexists with near-perfect design recognition (A-B) Mean repeated-cross-validation ROC-AUC for case-control discrimination (filled points) and for the most predictable recorded batch in a one-versus-rest analysis (open points) using frozen Geneformer donor means, HVG pseudobulk and PCA pseudobulk in GSE174188 (A) and GSE135779 (B). Horizontal segments connect the two tasks for each representation. (C) Matrix of disease AUC, best-batch AUC and their difference. Values are donor-level estimates; batch predictability documents design information retained by each representation but does not by itself identify that information as technical or biological.

Figure 3 Disease discrimination survives design restriction after controlling for lost sample size

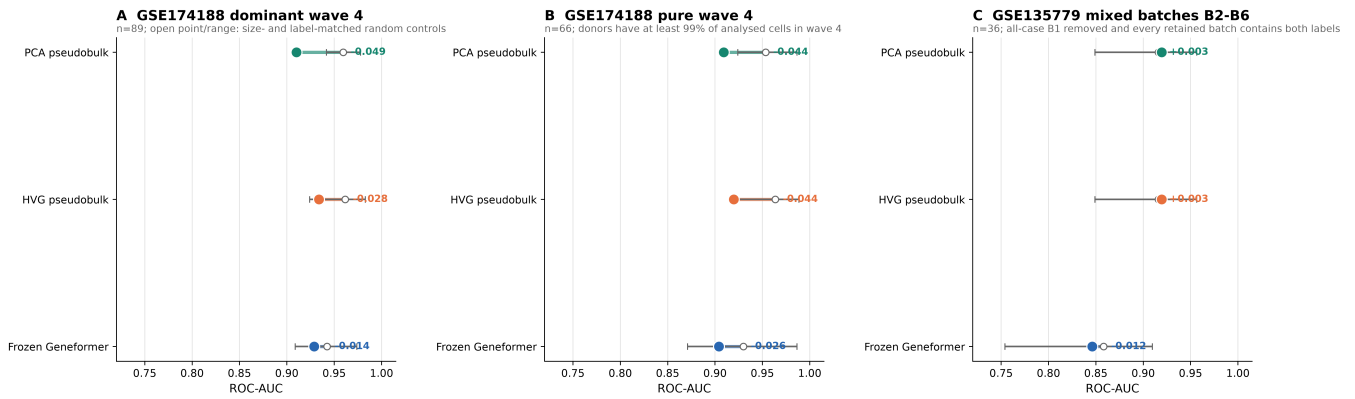


Figure 3. Disease discrimination survives design restriction after controlling for lost sample size (A) GSE174188 donors assigned to dominant processing wave 4 (n=89). (B) The stricter GSE174188 pure-wave-4 subset, requiring at least 99% of analysed cells in wave 4 (n=66). (C) GSE135779 donors from mixed-label batches B2-B6 after excluding the all-case B1 batch (n=36). Coloured points show mean repeated-cross-validation AUC in the observed stratum. Open points and grey bars show the mean and 2.5th-97.5th percentile from 20 random donor subsets matched to the observed donor count and case-control count. Labels report observed minus matched mean AUC. Restriction produced only small performance changes after accounting for the smaller analysis set.

Figure 4 Residualisation is a sensitivity operation, not a causal batch-effect estimate

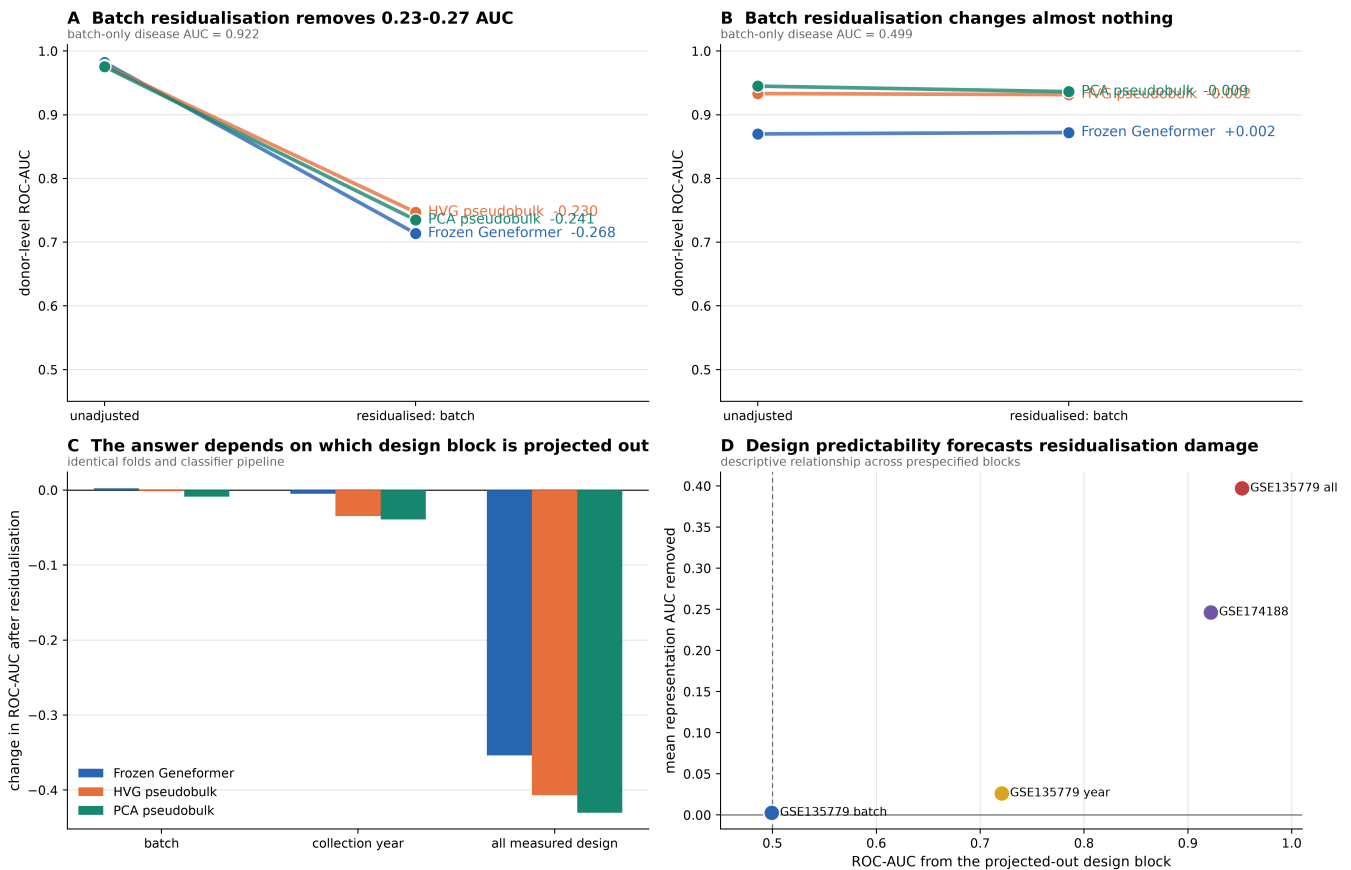


Figure 4. Residualisation is a sensitivity operation, not a causal batch-effect estimate (A) GSE174188 AUC before and after fold-contained residualisation on processing-wave fractions; the projected design block itself predicts disease at AUC 0.922. (B) GSE135779 AUC before and after residualisation on batch; batch alone is uninformative for disease at AUC 0.499. (C) Change in GSE135779 AUC after projecting out batch, collection year or the complete measured design block. (D) Across prespecified design blocks, the mean AUC removed from the molecular representations increases with the disease predictability of the projected block. All adjusted and unadjusted estimates use identical outer donor folds and training-only preprocessing. Residualisation therefore quantifies sensitivity to deleting design-aligned coordinates; under label-design collinearity it cannot identify a causal technical batch effect.

Figure 5 Generalisation, not internal separability, is where donor-level models break

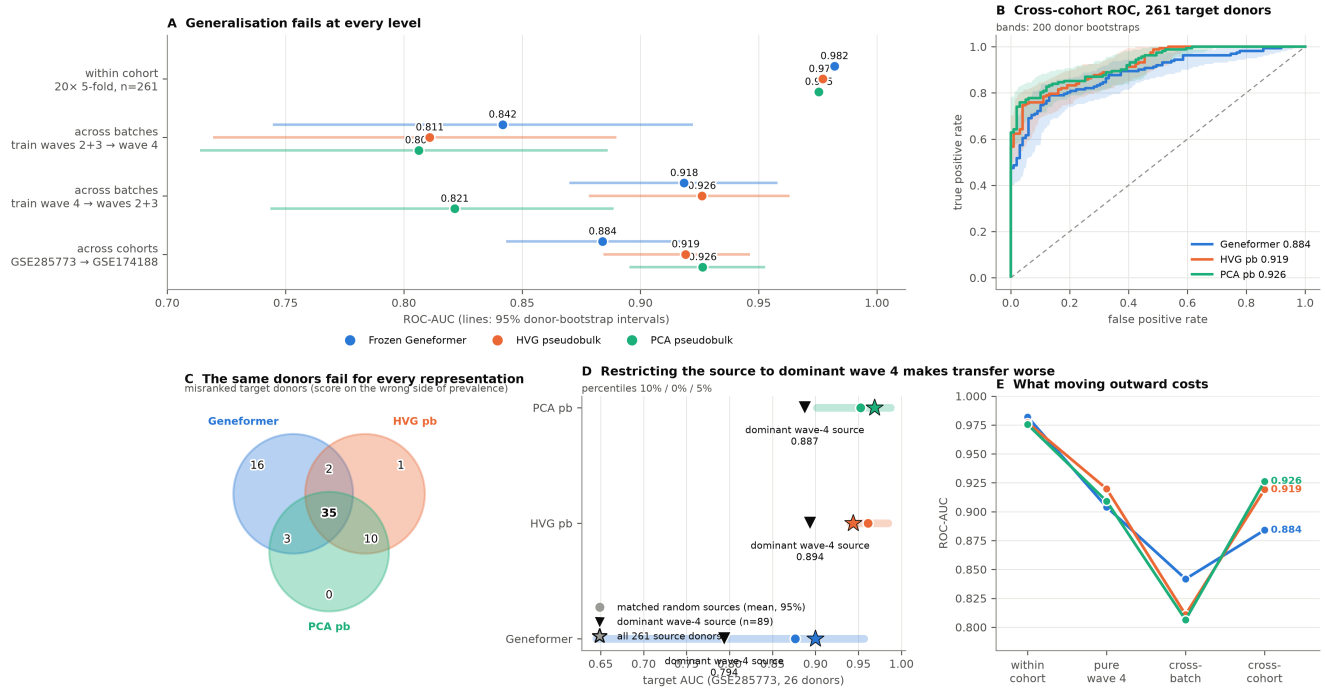


Figure 5. Generalisation, not internal separability, is where donor-level models break (A) Donor-level ROC-AUC for within-cohort evaluation, bidirectional cross-wave evaluation in GSE174188 and strict source-only transfer from GSE285773 to GSE174188. Horizontal lines are 95% donor-bootstrap intervals for transfer estimates; the within-cohort points summarize 20 repeated stratified five-fold analyses. (B) Target ROC curves for the 261-donor GSE174188 cohort after training on GSE285773, with pointwise bands from 200 stratified donor bootstrap resamples. (C) Overlap among target donors whose scores fell on the wrong side of target prevalence for the three representations, illustrating shared rather than method-specific failures. (D) Transfer to GSE285773 after restricting the GSE174188 source to dominant wave 4, compared with matched random source subsets and all 261 source donors. (E) AUC across the sequence from within-cohort evaluation to pure-wave, cross-batch and cross-cohort evaluation. Every transformation and classifier in cross-cohort analyses was fitted on source donors only.

Figure 6 Learned pooling does not rescue the frozen representation

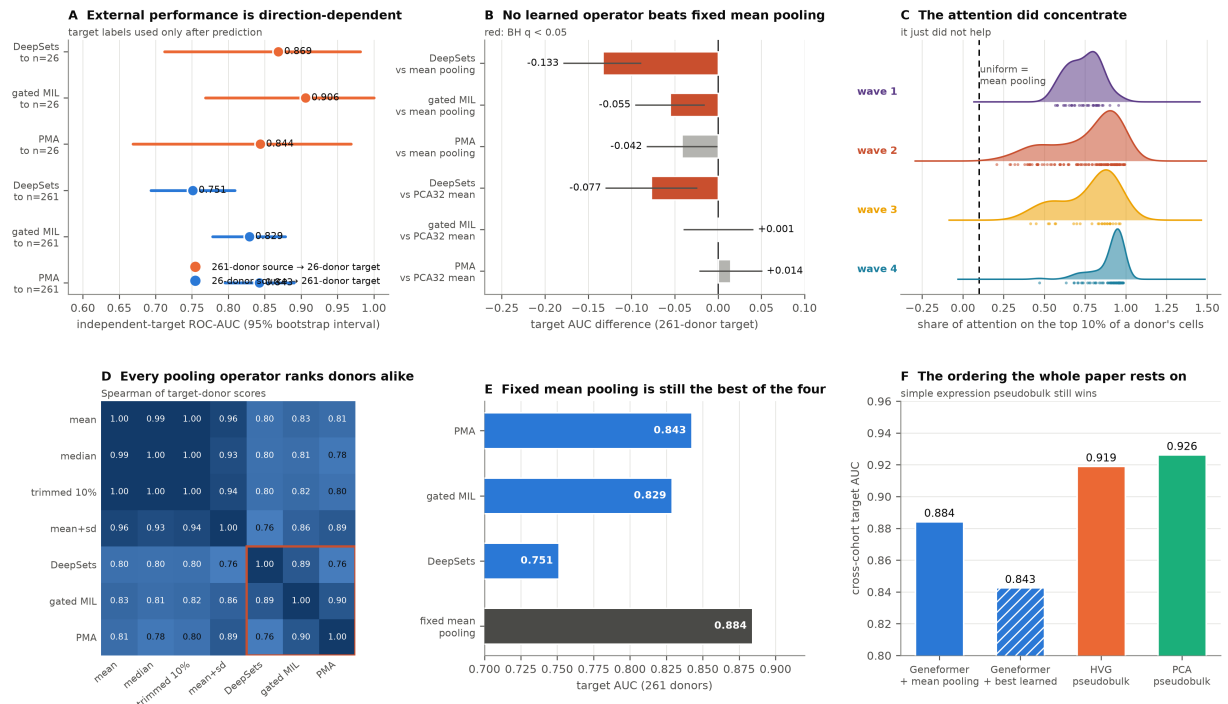


Figure 6. Learned pooling does not rescue the frozen representation (A) Independent-target AUC and 95% stratified donor-bootstrap intervals for DeepSets, gated-attention multiple-instance learning and pooling by multi-head attention in both transfer directions. Colours identify the source-target direction. (B) Paired target-donor AUC differences for the 261-donor target relative to the 1,152-dimensional frozen Geneformer donor mean and to a donor mean in the identical source-fitted PCA32 space. Red bars denote Benjamini-Hochberg-adjusted $q < 0.05$. (C) Distribution across source donors of the share of gated-attention mass assigned to the top 10% of cells, stratified by processing wave; the dashed line is the value under uniform attention. (D) Spearman correlations among independent-target donor scores from deterministic and learned pooling operators. (E) Target AUC for learned operators and fixed mean pooling in the 261-donor target. (F) Strict source-only cross-cohort AUC for fixed Geneformer mean pooling, the best learned operator and the two expression pseudobulk baselines. Learned pooling was selected using source donors only and did not recover the external performance gap.